

RAJATH R

Personal Data

Name: Rajath R
Place and Date of Birth: Bengaluru, India | 05 February 2000
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Education

Oct 2021 - Jun 2023	Master Of Science - Physics Fergusson College Pune, Maharashtra Grade: 8.0
Aug 2018 - Jul 2021	Bachelor Of Science - Physics, Mathematics & Computer Science Jain University Bengaluru, Karnataka Grade: 8.3

Research Experience

Jun 2024 - Sep 2024	REYES Program - University Of California, Berkeley - Collaborated in a research group under Dr. Martha Constantinou, Temple University. - Analyzed Lattice QCD simulation datasets to extract pion energy eigenvalues from two-point correlation functions across varying momenta. - Constructed and evaluated the pion's dispersion relation, comparing it with theoretical predictions to validate computational techniques and assess the accuracy of lattice QCD. - Presented findings to the REYES committee and co-authored a proceedings paper documenting the research.
Jan 2023 - Jul 2023	Dr. Rajeev S. Bhalerao (IISER Pune) - Research Intern - Worked on modelling nucleon structure functions using parton distribution function in FORTRAN. - Evolved the solved PDF's with QCD (DGLAP equations) and studied them for higher energy scales. - Involved extensive literature survey and thesis defence as a part of my final year masters.
Jan 2023 - March 2023	IceCube - Neutrinos in Deep Ice - Part of a team involved in a machine learning research code competition conducted by kaggle. - Worked with professors and industry professionals to understand neutrinos, the IceCube experiment and create ML models to refine the process of neutrino detection and train them with provided datasets.

Dec 2022 | Fergusson College Physics Exhibit

- Explained Quantum Field Theory using a simple model, to physics enthusiasts of all levels.
- Displayed simulations to explain virtual particles.
- Broke down and explained the meaning of Feynman Diagrams.

Sep 2019 - Feb 2020 | JNCASR Bengaluru - Undergraduate Research Student

- Conducted extensive literature review on superconductors and the theories behind their properties.
- Worked with a team of three and synthesised YBCO superconductor using multiple techniques, analyzed its properties and studied its magnetic behaviour.
- Presented a dissertation on our project and our findings at the presence of a panel of professors.

Publications

Debarshi Mukherjee, Dilan Arik, Navneeth C, Sufiyan Mirza, Ameer Mustafa, **Rajath R**, Anantha Srinivasan, Preksha Uniyal, Martha Constantinou - The Energy Spectrum of the Pion from Lattice QCD *arXiv* 2508.05721, 2025.

Skills

- C++, JAVA, PYTHON, FORTRAN, ROOT
- DATA ANALYSIS AND MACHINE LEARNING TECHNIQUES
- DEEP LEARNING FRAMEWORKS - TENSORFLOW, PYTORCH
- ORIGIN, MATLAB
- L^AT_EX
- SCIENTIFIC LITERATURE REVIEW AND PRESENTATION
- SPOKEN LANGUAGES : English, Kannada, Malayalam, Hindi, Tamil, Telugu, Marathi

Workshops & Seminars

- Workshop on Probability Theory by Dr. K.B. Athreya - Jain University, 2018
- 'Observing the Birth of the Universe' by Dr. Lyman Page - ICTS Bengaluru, 2019
- CERN India Week Exhibit & Lectures - Visvesvaraya Museum Bengaluru, 2019
- Workshop on Theory Of Computation - Mathematical Sciences Institute Belgaum, 2020
- Introduction to Lie Groups and Lie Algebra by Dr. Swarnamala Sirsi - Mysore University, 2021
- Symposium: Nucleon and Nuclear structure from Inclusive measurements - Jefferson Labs, 2023 (Online mode)
- Basics of Quantum Gravity - International Society for Quantum Gravity, 2023